

Norms and Singular Value Decomposition

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Norms

Vector Norms

We have used the *length* (= 2-norm) of vectors $\vec{x} \in \mathbb{R}^n$:

$$|\vec{x}| = \|\vec{x}\|_2 = \sqrt{\sum_{i=1}^n x_i^2} = \sqrt{\vec{x}^\top \vec{x}} = \sqrt{\vec{x} \cdot \vec{x}}$$

There is a more general definition of *a vector norm*: namely, any function $\|\cdot\| : \mathbb{R}^n \rightarrow \mathbb{R}$ that satisfies for all vectors $\vec{x}, \vec{y} \in \mathbb{R}^n$ and all scalars $r \in \mathbb{R}$:

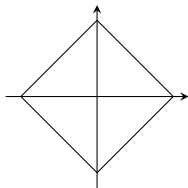
- 1) $\|\vec{x}\| \geq 0$, and $\|\vec{x}\| = 0$ if and only if $\vec{x} = \vec{0}$,
- 2) $\|\vec{x} + \vec{y}\| \leq \|\vec{x}\| + \|\vec{y}\|$,
- 3) $\|r\vec{x}\| = |r| \|\vec{x}\|$.

Often used are the *p-norms*, $p \geq 1$:

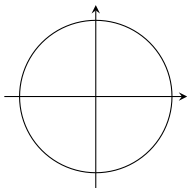
$$\|\vec{x}\|_p := \left(\sum_{i=1}^n |x_i|^p \right)^{1/p}$$

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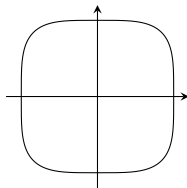
In $\mathbb{R}^2$, here are the unit balls, i.e., $\vec{x}$ with $\|\vec{x}\|_p = 1$:



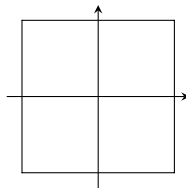
$p = 1$



$p = 2$



$p = 4$



$p = \infty$

$$\|\vec{x}\|_1 = \sum_{i=1}^n |x_i|$$

$$\|\vec{x}\|_\infty := \max_{1 \leq i \leq n} |x_i|$$

Induced Matrix Norms

A vector norm $\|\cdot\|$ induces (i.e., determines) a norm on matrices $\|\cdot\| : \mathbb{R}^{m \times n} \rightarrow \mathbb{R}$, the *induced matrix norm*:

$$\|A\| := \sup_{\vec{x} \in \mathbb{R}^n \setminus \{\vec{0}\}} \frac{\|A\vec{x}\|}{\|\vec{x}\|} = \sup_{\substack{\vec{x} \in \mathbb{R}^n \\ \|\vec{x}\|=1}} \|A\vec{x}\|$$

Intuitively, that is the largest effect that multiplication with A has on a vector relative to its norm. By the definition, for all $\vec{x}$:

$$\|A\vec{x}\| \leq \|A\| \|\vec{x}\|$$

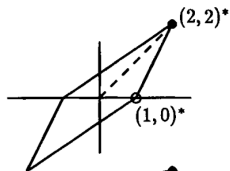
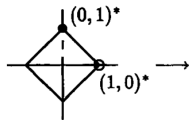
There is a notion of *a matrix norm* similar to the one for vectors. Not every matrix norm is induced by a vector norm. For example, here is the *Frobenius norm*:

$$\|A\|_F := \sqrt{\sum_{i=1}^m \sum_{j=1}^n |a_{ij}|^2}$$

Example for Induced Matrix Norms

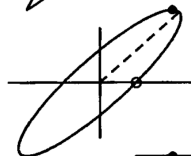
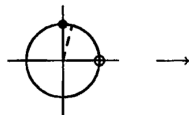
$$A = \begin{pmatrix} 1 & 2 \\ 0 & 2 \end{pmatrix}$$

1-norm:



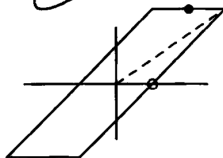
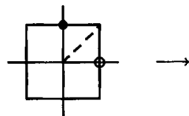
$$\|A\|_1 = 4$$

2-norm:



$$\|A\|_2 \approx 2.9208$$

∞ -norm:



$$\|A\|_\infty = 3$$

Properties

Lemma Let $\|\cdot\|$ denote an induced matrix norm. Then:

$$\|AB\| \leq \|A\| \cdot \|B\|$$

Proof $\|AB\vec{x}\| \leq \|A\| \|B\vec{x}\| \leq \|A\| \|B\| \|\vec{x}\|.$ QED

Lemma Let $Q \in \mathbb{R}^{m \times m}$ be an orthogonal matrix, and $A \in \mathbb{R}^{m \times n}$. Then:

$$\begin{aligned}\|Q\|_2 &= 1 \\ \|QA\|_2 &= \|A\|_2\end{aligned}$$

Proof We have $\|Q\vec{y}\|_2 = \|\vec{y}\|_2$ for all $\vec{y}$. QED

Singular Value Decomposition (SVD)

Definition

The *Singular Value Decomposition (SVD)* of a matrix $A \in \mathbb{R}^{m \times n}$ (where $m \geq n$) is a factorization

$$\boxed{A} = \boxed{U} \boxed{\Sigma} \boxed{V^T}$$

where $U \in \mathbb{R}^{m \times m}$ and $V \in \mathbb{R}^{n \times n}$ are orthogonal and $\Sigma \in \mathbb{R}^{m \times n}$ is diagonal. The diagonal entries $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_n \geq 0$ of Σ are called *singular values*. The (pairwise orthogonal) columns of U are the *left singular vectors*, and the (pairwise orthogonal) columns of V the *right singular vectors*.

The SVD describes the action of A as a composition of three linear maps:

- 1) rotation/reflection, as specified by V^T ; followed by
- 2) scaling along the axes, as specified by Σ ; followed by
- 3) another rotation/reflection, as specified by U .

One can show that every matrix A has an SVD!

$$\boxed{A} \quad \boxed{V} = \boxed{U} \quad \boxed{\Sigma}$$

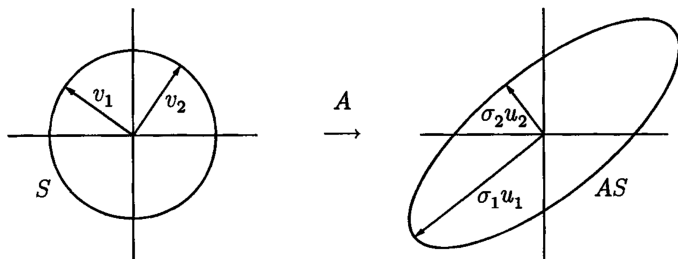
$$\boxed{A} \quad \left(\begin{array}{c|c|c} \vdots & & \vdots \\ \hline \vec{v}_1 & \cdots & \vec{v}_n \\ \hline \vdots & & \vdots \end{array} \right) = \left(\begin{array}{c|c|c} \vdots & & \vdots \\ \hline \vec{u}_1 & \cdots & \vec{u}_m \\ \hline \vdots & & \vdots \end{array} \right) \left(\begin{array}{ccc} \sigma_1 & & \\ & \ddots & \\ & & \sigma_n \end{array} \right)$$

$$\boxed{A} \quad \left(\begin{array}{c|c|c} \vdots & & \vdots \\ \hline \vec{v}_1 & \cdots & \vec{v}_n \\ \hline \vdots & & \vdots \end{array} \right) = \left(\begin{array}{c|c|c} \vdots & & \vdots \\ \hline \vec{u}_1 & \cdots & \vec{u}_n \\ \hline \vdots & & \vdots \end{array} \right) \left(\begin{array}{ccc} \sigma_1 & & \\ & \ddots & \\ & & \sigma_n \end{array} \right)$$

$$A \quad \vec{v}_i = \sigma_i \vec{u}_i \quad (1 \leq i \leq n)$$

These n equations tell us how A acts on the right (!) singular vectors $\vec{v}_i$.

The image of a unit sphere under any matrix is a hyperellipse.



$$A\vec{x} = \vec{b}$$

Represent $\vec{x}$ in the basis of the right singular vectors: $\vec{x}' = V^T \vec{x}$

Represent $\vec{b}$ in the basis of the left singular vectors: $\vec{b}' = U^T \vec{b}$

Then:

$$\vec{b}' = U^T \vec{b} = U^T A \vec{x} = U^T U \Sigma V^T \vec{x} = \Sigma V^T \vec{x} = \Sigma \vec{x}'$$

That is, the map defined by A is the map of a diagonal matrix, if domain and codomain bases consist of the singular vectors.

Existence, Uniqueness and Computation

Theorem Every matrix $A \in \mathbb{R}^{m \times n}$ has an SVD. The singular values σ_i are uniquely determined. Further, if A is square and the σ_i are distinct, the left and right singular vectors are uniquely determined up to signs.

Proof See, e.g., [Trefethen, Bau].

One can compute (an approximation of) an SVD with $O(mn^2)$ flops in practice. This needs to involve an iterative phase, as the singular values are in general irrational (like eigenvalues). This phase takes infinitely many iterations in theory, but is not a bottleneck in practice.

SVD vs. Eigenvalue Decomposition

Recall that diagonalizable matrices have an eigenvalue decomposition:

$$A = U\Sigma U^{-1} \quad \Longleftrightarrow \quad AU = U\Sigma$$

The columns of U are the eigenvectors of A .

Differences between eigenvalue decomposition and SVD:

- ▶ To “make” A diagonal using the eigenvalue decomposition, the same basis for domain and range can be used (consisting of the eigenvectors).
- ▶ The eigenvectors are in general not orthogonal; the singular vectors are.
- ▶ Not every matrix A is diagonalizable, not even square matrices. In contrast, the SVD always exists.
- ▶ In applications, eigenvalues tend to be useful for the iterated behaviour of A , such as matrix powers A^k . The SVD is useful for the analysis of A itself.

Matrix Properties via SVD

Let $r \leq \min\{m, n\}$ be the number of nonzero singular values of $A \in \mathbb{R}^{m,n}$.
We have $\sigma_1 \geq \dots \geq \sigma_r > 0$.

Theorem The rank of A is r .

Proof The rank of a diagonal matrix is equal to the number of its nonzero entries, and in the decomposition $A = U\Sigma V^\top$ the matrices U and V have full rank. Therefore $\text{rank}(A) = \text{rank}(\Sigma) = r$. QED

This is useful for practical computation of the rank (choose a tolerance).

Theorem $\langle \vec{u}_1, \dots, \vec{u}_r \rangle$ is an orthogonal basis of A 's column space. Similarly, $\langle \vec{v}_{r+1}, \dots, \vec{v}_n \rangle$ is an orthogonal basis of A 's nullspace.

Proof Follows from the facts that $\langle \vec{e}_1, \dots, \vec{e}_r \rangle$ is a basis of Σ 's column space and $\langle \vec{e}_{r+1}, \dots, \vec{e}_n \rangle$ is a basis of Σ 's nullspace. QED

In practice, column space and nullspace can also be computed via QR factorization, but it is more accurate via SVD.

Theorem $\|A\|_2 = \sigma_1$.

Proof $\|A\|_2 = \|\mathbf{U}\Sigma\mathbf{V}^\top\|_2 = \|\Sigma\|_2 = \sigma_1$.

QED

This represents the standard method of computing $\|A\|_2$.

Theorem The nonzero singular values of A are the square roots of the nonzero eigenvalues of $AA^\top$.

Proof We have:

$$AA^\top = \mathbf{U}\Sigma\mathbf{V}^\top\mathbf{V}\Sigma^\top\mathbf{U}^\top = \mathbf{U}\Sigma\Sigma^\top\mathbf{U}^\top$$

So the matrices $AA^\top$ and $\Sigma\Sigma^\top$ are similar, and hence have the same eigenvalues. The nonzero eigenvalues of $\Sigma\Sigma^\top$ are $\sigma_1^2, \dots, \sigma_r^2$.

QED

Theorem Let $A \in \mathbb{R}^{n \times n}$ be a square matrix. Then $|\det(A)| = \prod_{i=1}^n \sigma_i$.

Proof The determinant of any orthogonal matrix Q is 1 in absolute value:

$$\det(Q)^2 = \det(Q)\det(Q^\top) = \det(QQ^\top) = \det(I) = 1$$

Hence:

$$|\det(A)| = |\det(\mathbf{U}\Sigma\mathbf{V}^\top)| = \underbrace{|\det(\mathbf{U})|}_{1} |\det(\Sigma)| \underbrace{|\det(\mathbf{V}^\top)|}_{1} = \prod_{i=1}^n \sigma_i \quad \text{QED}$$

Low-Rank Approximation

Theorem For any $k \leq r$, define $A_k := \sum_{i=1}^k \sigma_i \vec{u}_i \vec{v}_i^T$. Then

$$\|A - A_k\|_2 = \inf_{\substack{B \in \mathbb{R}^{m \times n} \\ \text{rank}(B) \leq k}} \|A - B\|_2 = \sigma_{k+1} \quad (:= 0 \text{ if } k = \min\{m, n\}).$$

Proof We have:

$$A_k = U \begin{pmatrix} \sigma_1 & & & & \\ & \ddots & & & \\ & & \sigma_k & & \\ & & & 0 & \\ & & & & \ddots \end{pmatrix} V^T$$

So $\|A - A_k\|_2 = \sigma_{k+1}$. Suppose $B \in \mathbb{R}^{m \times n}$ has $\text{rank}(B) \leq k$ and $\|A - B\|_2 < \sigma_{k+1}$. By rank-nullity, the nullspace of B , say W , is at least $(n - k)$ -dimensional. For all $\vec{w} \in W \setminus \{\vec{0}\}$:

$$\|A\vec{w}\|_2 = \|(A - B)\vec{w}\|_2 \leq \|A - B\|_2 \|\vec{w}\|_2 < \sigma_{k+1} \|\vec{w}\|_2$$

But there is a $(k + 1)$ -dimensional subspace of $\mathbb{R}^n$ with $\|A\vec{w}\|_2 \geq \sigma_{k+1} \|\vec{w}\|_2$, namely the span of first $k + 1$ right singular vectors $\vec{v}_1, \dots, \vec{v}_{k+1}$. This vector space overlaps nontrivially with W , a contradiction. QED